

Infinitesimal Calculus 2

Master Theorems & Reference Sheet
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1 Improper Integrals (Lecture 1)

Definition 1.1 (Improper integrals). • **Type I** (infinite limit): $\int_a^\infty f(x) dx = \lim_{b \rightarrow \infty} \int_a^b f(x) dx$.

• **Type II** (unbounded integrand at endpoint b): $\int_a^b f(x) dx = \lim_{\varepsilon \rightarrow 0^+} \int_a^{b-\varepsilon} f(x) dx$.

The integral *converges* if the limit exists and is finite; otherwise it *diverges*.

Reference integrals (p-test)

Integral	Converges when	Diverges when
$\int_1^\infty \frac{dx}{x^p}$	$p > 1$	$p \leq 1$
$\int_0^1 \frac{dx}{x^p}$	$p < 1$	$p \geq 1$

Convergence tests (for $f \geq 0$)

Theorem 1.2 (Comparison Test). Suppose $0 \leq f(x) \leq g(x)$ for all large x .

- If $\int g$ converges, then $\int f$ converges.
- If $\int f$ diverges, then $\int g$ diverges.

When to use: When you can bound f from above/below by a known integral.

Theorem 1.3 (Limit Comparison Test). Let $f, g \geq 0$ and $L = \lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} \in (0, \infty)$. Then $\int_a^\infty f$ and $\int_a^\infty g$ have the same convergence behaviour. *When to use:* When $f \sim C \cdot g$ asymptotically.

Remark 1.4 (Asymptotic equivalences near 0). As $\alpha \rightarrow 0$:

$$\frac{\sin \alpha}{\alpha} \rightarrow 1, \quad \frac{e^\alpha - 1}{\alpha} \rightarrow 1, \quad \frac{\ln(1 + \alpha)}{\alpha} \rightarrow 1.$$

Strategy: Near a problematic point, replace each factor by its asymptotic equivalent, then compare with a p -integral.

2 Topology of $\mathbb{R}^n$ (Lectures 1 & 3)

Definition 2.1 (Euclidean space $\mathbb{R}^n$). $\mathbb{R}^n$ consists of vectors $\bar{x} = (x_1, \dots, x_n)$ with the **Euclidean norm**

$$\|\bar{x}\|_2 = \sqrt{x_1^2 + \dots + x_n^2}.$$

Other common norms: $\|\bar{x}\|_1 = \sum |x_i|$, $\|\bar{x}\|_\infty = \max_i |x_i|$.

Definition 2.2 (Norm axioms). $\|\cdot\| : \mathbb{R}^n \rightarrow \mathbb{R}^+$ is a norm if:

1. $\|\bar{x}\| = 0 \iff \bar{x} = 0$,

2. $\|\alpha\bar{x}\| = |\alpha| \|\bar{x}\|$,
3. $\|\bar{x} + \bar{y}\| \leq \|\bar{x}\| + \|\bar{y}\|$ (triangle inequality).

Theorem 2.3 (Cauchy–Schwarz Inequality).

$$\left(\sum_{i=1}^n |x_i y_i| \right)^2 \leq \left(\sum_{i=1}^n x_i^2 \right) \left(\sum_{i=1}^n y_i^2 \right).$$

Proof idea: Define $P(t) = \sum (t|x_i| - |y_i|)^2 \geq 0$; this quadratic in t has non-positive discriminant.

Definition 2.4 (Metric / distance). $d(\bar{x}, \bar{y}) = \|\bar{x} - \bar{y}\|$. It satisfies: $d \geq 0$; $d = 0 \iff \bar{x} = \bar{y}$; symmetry; triangle inequality.

Topological notions

Definition 2.5 (Open ball). $B(\bar{a}, r) = \{\bar{x} : \|\bar{x} - \bar{a}\| < r\}$.

Let $E \subseteq \mathbb{R}^n$.

Type	Condition	Notation
Interior point	$\exists r > 0 : B(\bar{a}, r) \subseteq E$	$\text{Int } E$
Exterior point	$\exists r > 0 : B(\bar{a}, r) \cap E = \emptyset$	$\text{Ext } E$
Boundary point	Every $B(\bar{a}, r)$ meets both E and $\mathbb{R}^n \setminus E$	∂E

Definition 2.6 (Open, closed, bounded, compact). • **Open:** $E = \text{Int } E$ (every point is interior).

- **Closed:** $\partial E \subseteq E$ (equivalently: $\mathbb{R}^n \setminus E$ is open).
- **Bounded:** $E \subseteq B(\bar{a}, r)$ for some ball.
- **Compact:** closed + bounded.

Theorem 2.7 (Weierstrass — existence of extrema). *If f is continuous on a compact set $E \subseteq \mathbb{R}^n$, then:*

1. f is bounded on E .
2. f attains its supremum $M = \sup_E f$ and infimum $m = \inf_E f$ on E .

3 Multivariable Functions — Limits & Continuity (Lectures 2–3)

Definition 3.1 (Limit — Heine definition). $\lim_{\bar{x} \rightarrow \bar{a}} f(\bar{x}) = L$ if for every sequence $\bar{x}^{(m)} \rightarrow \bar{a}$ (with $\bar{x}^{(m)} \neq \bar{a}$), we have $f(\bar{x}^{(m)}) \rightarrow L$.

Definition 3.2 (Limit — Cauchy / ε - δ definition). $\lim_{\bar{x} \rightarrow \bar{a}} f(\bar{x}) = L$ if

$$\forall \varepsilon > 0 \exists \delta > 0 : 0 < \|\bar{x} - \bar{a}\| < \delta \implies |f(\bar{x}) - L| < \varepsilon.$$

Theorem 3.3. *The Heine and Cauchy definitions are equivalent.*

Techniques for evaluating limits

Remark 3.4 (Two-path test for non-existence). To prove $\lim f$ **does not exist**: find two paths to $\bar{a}$ giving different limit values.

Remark 3.5 (Polar coordinates). Set $x = \rho \cos \theta$, $y = \rho \sin \theta$. If $f(\rho, \theta) \rightarrow L$ as $\rho \rightarrow 0$ uniformly in θ , then $\lim_{(x,y) \rightarrow (0,0)} f = L$.

Pitfall: If the expression in polar still depends on θ without vanishing, the limit may not exist; fall back to the two-path method.

Remark 3.6 (Squeeze theorem). If $|f(\bar{x}) - L| \leq g(\bar{x})$ and $g(\bar{x}) \rightarrow 0$, then $f(\bar{x}) \rightarrow L$.

Key examples

Function at $(0,0)$	Result	Method
$\frac{xy}{x^2 + y^2}$	DNE	Paths $y = 0$ (value 0), $y = x$ (value $\frac{1}{2}$)
$\frac{xy^2}{x^2 + y^2}$	0	Polar: $\rho \cos \theta \sin^2 \theta \rightarrow 0$; squeeze $\leq y $
$\frac{xy^2}{x^2 + y^4}$	DNE	Paths $y = 0$ (value 0), $x = y^2$ (value $\frac{1}{2}$)
$\frac{x^2 y^2 (x^2 + y^2)}{x^4 + y^4}$	0	Polar, bound $\cos^4 \theta + \sin^4 \theta \geq \frac{1}{2}$
$\frac{\sin(x-y)}{x-y}$	DNE	Not defined on $y = x$; no punctured nbhd

Remark 3.7 (Useful bound in polar).

$$\cos^4 \theta + \sin^4 \theta = 1 - 2 \cos^2 \theta \sin^2 \theta = 1 - \frac{1}{2} \sin^2 2\theta \geq \frac{1}{2}.$$

Definition 3.8 (Continuity). f is **continuous** at $\bar{a}$ if $\lim_{\bar{x} \rightarrow \bar{a}} f(\bar{x}) = f(\bar{a})$. Equivalently: $\Delta f(\bar{a}) = f(\bar{a} + \Delta \bar{x}) - f(\bar{a}) \rightarrow 0$ as $\Delta \bar{x} \rightarrow 0$.

4 Partial Derivatives & Differentiability (Lecture 4)

Partial derivatives

Definition 4.1 (First-order partial derivatives). For $z = f(x, y)$ at (x_0, y_0) :

$$f'_x(x_0, y_0) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x, y_0) - f(x_0, y_0)}{\Delta x}, \quad f'_y(x_0, y_0) = \lim_{\Delta y \rightarrow 0} \frac{f(x_0, y_0 + \Delta y) - f(x_0, y_0)}{\Delta y}.$$

Computationally: differentiate w.r.t. x treating y constant (and vice versa).

Differentiability

Definition 4.2 (Differentiability). $f(x, y)$ is **differentiable** at (x_0, y_0) if the total increment satisfies

$$\Delta f = A \Delta x + B \Delta y + \alpha \Delta x + \beta \Delta y,$$

where $\alpha, \beta \rightarrow 0$ as $(\Delta x, \Delta y) \rightarrow (0, 0)$. The expression $A \Delta x + B \Delta y$ is the **total differential** df . Equivalently:

$$\lim_{\Delta \rho \rightarrow 0} \frac{\varepsilon}{\Delta \rho} = 0, \quad \varepsilon = \Delta f - f_x \Delta x - f_y \Delta y, \quad \Delta \rho = \sqrt{\Delta x^2 + \Delta y^2}.$$

Remark 4.3 (Intuition for $\varepsilon/R \rightarrow 0$). Δf is the *actual* change in f ; $df = f_x \Delta x + f_y \Delta y$ is the *linear approximation* of that change. Their difference ε is the error. Differentiability asks:

Does the error ε shrink to zero faster than the distance R moved?

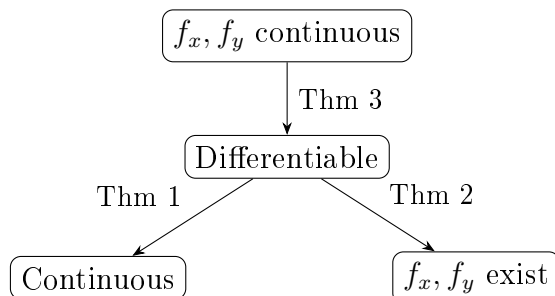
A small absolute error is not enough — ε must become *negligible relative to R* (i.e. $\varepsilon = o(R)$). Geometrically: the tangent plane is a *good* local approximation of f , not just any approximation.

Hierarchy of properties

Theorem 4.4 (Thm 1: Differentiable $\Rightarrow$ Continuous). *If f is differentiable at (x_0, y_0) , it is continuous there. (Converse is false.)*

Theorem 4.5 (Thm 2: Differentiable $\Rightarrow$ partial derivatives exist). *If f is differentiable at (x_0, y_0) , then f_x and f_y exist there, and $A = f_x$, $B = f_y$. (Converse is false.)*

Theorem 4.6 (Thm 3 — Sufficient condition for differentiability). *If f_x and f_y exist in a neighbourhood of (x_0, y_0) and are continuous there, then f is differentiable at (x_0, y_0) . (Converse is false.)*



Warning 4.7. None of the reverse implications hold in general!

Counterexamples (know these!)

Function	At $(0, 0)$	Holds	Fails
$f = \frac{xy}{x^2 + y^2}, f(0, 0) = 0$	$(0, 0)$	$f_x = f_y = 0$ exist	Continuous, differentiable
$f = \frac{x^2 y}{x^4 + y^2}, f(0, 0) = 0$	$(0, 0)$	$f_x = f_y = 0$ exist	Continuous, differentiable
$f = x + \sqrt[3]{y^2}$	$(0, 0)$	Continuous	Partial derivatives
$f = (x^2 + y^2) \sin \frac{1}{x^2 + y^2}, f(0, 0) = 0$	$(0, 0)$	Differentiable ($df = 0$)	f_x, f_y continuous

How to check differentiability at (x_0, y_0)

Step 1 — Quick pass (sufficient condition): If f_x and f_y both exist and are *continuous* near the point $\Rightarrow$ differentiable. Done.

Step 2 — Must use definition (piecewise / suspicious point):

1. Compute $f_x(x_0, y_0)$ and $f_y(x_0, y_0)$ from the *limit definition* (not formula).
2. Form the error: $\varepsilon = \Delta f - f_x \Delta x - f_y \Delta y$.
3. Check: $\lim_{R \rightarrow 0} \frac{\varepsilon}{R} = 0$ where $R = \sqrt{\Delta x^2 + \Delta y^2}$.
4. *Trick:* substitute polar coords $\Delta x = r \cos \theta$, $\Delta y = r \sin \theta$, $R = r$. If after simplification r cancels and a θ -dependent expression remains $\Rightarrow$ limit does not exist $\Rightarrow$ **not differentiable**.

Step 3 — Consequences (check the chain):

$$f_x, f_y \text{ continuous} \Rightarrow \text{differentiable} \Rightarrow \text{continuous} \Rightarrow \text{partials exist}$$

None of the reverse implications hold.

Warning — directional derivatives: $D_{\hat{u}}f = \nabla f \cdot \hat{u}$ is valid *only if f is differentiable*. At suspicious points, always use the definition: $D_{\hat{u}}f(x_0, y_0) = \lim_{t \rightarrow 0} \frac{f(x_0 + tu_1, y_0 + tu_2) - f(x_0, y_0)}{t}$.

Total differential & linear approximation

Definition 4.8 (Total differential).

$$df = f_x dx + f_y dy \quad (\text{or } f_x \Delta x + f_y \Delta y \text{ for finite increments}).$$

Remark 4.9 (Linear approximation).

$$f(x_0 + \Delta x, y_0 + \Delta y) \approx f(x_0, y_0) + f_x(x_0, y_0) \Delta x + f_y(x_0, y_0) \Delta y.$$

Use near a known point (x_0, y_0) where f is differentiable.

Remark 4.10 (Tangent plane to $z = f(x, y)$).

$$z - f(a, b) = f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

Normal direction: $\bar{n} = (f_x(a, b), f_y(a, b), -1)$.

Higher-order partial derivatives

Definition 4.11. $f_{xx} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right)$, $f_{xy} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right)$, etc.

Theorem 4.12 (Schwarz / Clairaut — mixed partial derivatives). *If f_{xy} and f_{yx} are both continuous in a neighbourhood of (x_0, y_0) , then*

$$f_{xy}(x_0, y_0) = f_{yx}(x_0, y_0).$$

5 Directional Derivative & Gradient (Week 3)

Definition 5.1 (Directional derivative). For a unit vector $\bar{u}$:

$$D_{\bar{u}}f(\bar{a}) = \lim_{t \rightarrow 0} \frac{f(\bar{a} + t\bar{u}) - f(\bar{a})}{t}.$$

If f is differentiable: $D_{\bar{u}}f = \nabla f \cdot \bar{u}$.

Definition 5.2 (Gradient).

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right) \quad (\text{or with } \frac{\partial f}{\partial z} \text{ in } \mathbb{R}^3).$$

Theorem 5.3 (Properties of the gradient). • $\|\nabla f\| = \text{maximum rate of change of } f$.

- The direction of ∇f gives the direction of steepest ascent.
- $\nabla f \perp$ level curves (level surfaces).

Remark 5.4 (Tangent plane to level surface $F(x, y, z) = C$).

$$F_x(P)(x - x_0) + F_y(P)(y - y_0) + F_z(P)(z - z_0) = 0, \quad P = (x_0, y_0, z_0).$$

6 Chain Rule (Weeks 2–4)

Theorem 6.1 (Chain rule — one parameter). If $z = f(x, y)$, $x = x(t)$, $y = y(t)$:

$$\frac{dz}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}.$$

Theorem 6.2 (Chain rule — two parameters). If $z = f(x, y)$, $x = x(u, v)$, $y = y(u, v)$:

$$\frac{\partial z}{\partial u} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial u}, \quad \frac{\partial z}{\partial v} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial v}.$$

Remark 6.3 (Polar chain rule). $w = f(x, y)$ with $x = r \cos t$, $y = r \sin t$:

$$\frac{1}{r} \frac{\partial w}{\partial t} = -f_x \sin t + f_y \cos t.$$

Fundamental Theorem of Calculus — applied to partial derivatives

Theorem 6.4 (FTC with variable limits). If $f(t)$ is continuous, define $F(x, y) = \int_x^y f(t) dt$.

Then:

$$\frac{\partial F}{\partial y} = f(y), \quad \frac{\partial F}{\partial x} = -f(x).$$

More generally, if both limits depend on parameters:

$$\frac{\partial}{\partial u} \int_{a(u)}^{b(u)} f(t) dt = f(b(u)) \cdot b'(u) - f(a(u)) \cdot a'(u).$$

Remark 6.5 (Why the minus sign on the lower limit). Split at any fixed point c :

$$\int_x^y f(t) dt = \int_c^y f(t) dt - \int_c^x f(t) dt.$$

Differentiating the second term w.r.t. x gives $-f(x)$.

Remark 6.6 (Direct usage — do NOT integrate). When asked to differentiate $z = \int_x^y f(t) dt$, apply FTC immediately. There is no need (and often no way) to find an antiderivative first.

Classic trap: $f(t) = e^{t^2}$ has *no elementary antiderivative*. Integration by parts will not close. Apply FTC directly:

$$\frac{\partial z}{\partial y} = e^{y^2}, \quad \frac{\partial z}{\partial x} = -e^{x^2}.$$

7 Taylor Polynomial & Local Extrema (Week 4)

Theorem 7.1 (Taylor polynomial of order 2).

$$f(a+h, b+k) = f(a, b) + [f_x h + f_y k] + \frac{1}{2} [f_{xx} h^2 + 2f_{xy} hk + f_{yy} k^2] + \dots$$

Theorem 7.2 (Second derivative test for local extrema). Let (a, b) be a critical point ($\nabla f = 0$). Define the **Hessian determinant**:

$$D = f_{xx}f_{yy} - (f_{xy})^2.$$

D	f_{xx}	Conclusion
$D > 0$	$f_{xx} > 0$	Local minimum
$D > 0$	$f_{xx} < 0$	Local maximum
$D < 0$	—	Saddle point
$D = 0$	—	Inconclusive

Theorem 7.3 (Sylvester's criterion (n variables)). For the Hessian matrix H with leading principal minors $\Delta_1, \Delta_2, \dots, \Delta_n$:

- **Positive definite** (local min): all $\Delta_k > 0$.
- **Negative definite** (local max): signs alternate $\Delta_1 < 0, \Delta_2 > 0, \Delta_3 < 0, \dots$
- Otherwise: indefinite (saddle) or inconclusive.

Remark 7.4 (Positive definite quadratic form in 2 variables). $Q(h, k) = ah^2 + 2bhk + ck^2$:

- Positive definite $\iff a > 0$ and $ac - b^2 > 0$.
- Negative definite $\iff a < 0$ and $ac - b^2 > 0$.

8 Implicit Functions & Lagrange Multipliers (Week 5)

Theorem 8.1 (Implicit Function Theorem (2 variables)). If $F(x, y) = 0$, F is C^1 , $F(x_0, y_0) = 0$, and $F_y(x_0, y_0) \neq 0$, then near (x_0, y_0) there is a unique function $y = \varphi(x)$ with $F(x, \varphi(x)) = 0$ and

$$\varphi'(x) = -\frac{F_x}{F_y}.$$

Theorem 8.2 (Implicit Function Theorem (3 variables)). For $F(x, y, z) = 0$ with $F_z \neq 0$:

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z}, \quad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z}.$$

Theorem 8.3 (Lagrange Multipliers). To optimize $f(x, y)$ subject to $g(x, y) = 0$, solve:

$$\nabla f = \lambda \nabla g, \quad g(x, y) = 0.$$

i.e. $f_x = \lambda g_x$, $f_y = \lambda g_y$, $g = 0$.

For **two constraints** $g_1 = 0$, $g_2 = 0$: $\nabla f = \lambda_1 \nabla g_1 + \lambda_2 \nabla g_2$.

Remark 8.4 (Global extrema on compact domain). 1. Find interior critical points ($\nabla f = 0$).

2. Find boundary extrema (parametrize boundary or use Lagrange multipliers).

3. Compare all candidate values; largest = global max, smallest = global min.

9 Double Integrals (Weeks 6–7)

Definition 9.1 (Double integral properties). $\iint_D f \, dA$: linearity, additivity over regions, monotonicity ($f \geq 0 \Rightarrow \iint f \geq 0$).

Theorem 9.2 (Fubini's Theorem). If f is continuous on D :

- **Type I:** $D = \{a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$

$$\iint_D f \, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) \, dy \, dx.$$

- **Type II:** $D = \{c \leq y \leq d, h_1(y) \leq x \leq h_2(y)\}$

$$\iint_D f \, dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) \, dx \, dy.$$

Theorem 9.3 (Change of variables in double integrals).

$$\iint_D f(x, y) \, dA = \iint_{D^*} f(x(u, v), y(u, v)) |J| \, du \, dv,$$

where $J = \frac{\partial(x, y)}{\partial(u, v)} = x_u y_v - x_v y_u$ is the **Jacobian**.

Remark 9.4 (Polar coordinates: $x = r \cos \theta$, $y = r \sin \theta$). $|J| = r$, so:

$$\iint_D f(x, y) \, dx \, dy = \iint_{D^*} f(r \cos \theta, r \sin \theta) \cdot r \, dr \, d\theta.$$

Use when: $x^2 + y^2$ appears or the domain is circular/sector-shaped.

10 Line Integrals (Weeks 7–8)

Definition 10.1 (Line integral of Type I — scalar / arc length).

$$\int_C f \, ds = \int_a^b f(\mathbf{r}(t)) |\mathbf{r}'(t)| \, dt, \quad |\mathbf{r}'(t)| = \sqrt{x'(t)^2 + y'(t)^2}.$$

If $f = 1$: gives the arc length of C .

Definition 10.2 (Line integral of Type II — vector field / work).

$$\int_C P \, dx + Q \, dy = \int_a^b [P(\mathbf{r}(t))x'(t) + Q(\mathbf{r}(t))y'(t)] \, dt.$$

Note: reversing the orientation negates the integral.

11 Green's Theorem & Conservative Fields (Week 8)

Theorem 11.1 (Green's Theorem). *Let C be a positively oriented (CCW), simple, closed curve bounding D :*

$$\oint_C P \, dx + Q \, dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA.$$

Applications: area of $D = \oint_C x \, dy = - \oint_C y \, dx = \frac{1}{2} \oint_C (x \, dy - y \, dx)$.

Theorem 11.2 (Conservative (gradient) fields — equivalent conditions). *For $\mathbf{F} = (P, Q)$ on a simply connected domain, the following are equivalent:*

1. $\mathbf{F} = \nabla \varphi$ for some potential φ (conservative).
2. $\int_C \mathbf{F} \cdot d\mathbf{r}$ is path-independent.
3. $\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$ for every closed curve.
4. $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$ (necessary; sufficient on simply connected domain).

If conservative: $\int_C \mathbf{F} \cdot d\mathbf{r} = \varphi(\text{end}) - \varphi(\text{start})$.

Remark 11.3 (Finding the potential). 1. Integrate: $\varphi = \int P \, dx + g(y)$.

2. Differentiate: $\varphi_y = Q$, solve for $g(y)$.

12 Triple Integrals (Weeks 8–9)

Theorem 12.1 (Fubini for triple integrals). $\iiint_B f \, dV = \iiint f(x, y, z) \, dx \, dy \, dz$ (iterate in any order over rectangular box). *For general domains: set up variable bounds from a sketch.*

Remark 12.2 (Cylindrical coordinates: $x = r \cos \theta$, $y = r \sin \theta$, $z = z$). $|J| = r$:

$$\iiint f dV = \iiint f(r \cos \theta, r \sin \theta, z) \cdot r dr d\theta dz.$$

Use when: $x^2 + y^2$ appears (circular symmetry in xy -plane).

Remark 12.3 (Spherical coordinates: $x = \rho \sin \varphi \cos \theta$, $y = \rho \sin \varphi \sin \theta$, $z = \rho \cos \varphi$). $|J| = \rho^2 \sin \varphi$, with $\rho \geq 0$, $0 \leq \varphi \leq \pi$, $0 \leq \theta \leq 2\pi$:

$$\iiint f dV = \iiint f(\dots) \cdot \rho^2 \sin \varphi d\rho d\varphi d\theta.$$

Use when: $x^2 + y^2 + z^2$ appears (spherical symmetry).

13 Surface Integrals (Week 10)

Definition 13.1 (Surface integral of Type I — scalar).

$$\iint_S f dS = \iint_D f(\mathbf{r}(u, v)) |\mathbf{r}_u \times \mathbf{r}_v| du dv.$$

For $z = g(x, y)$: $dS = \sqrt{1 + g_x^2 + g_y^2} dx dy$.

Definition 13.2 (Surface integral of Type II — flux).

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_D \mathbf{F} \cdot (\mathbf{r}_u \times \mathbf{r}_v) du dv.$$

Definition 13.3 (Divergence and curl).

$$\operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z},$$

$$\operatorname{curl} \mathbf{F} = \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}.$$

14 Gauss & Stokes Theorems (Week 11)

Theorem 14.1 (Divergence Theorem (Gauss)).

$$\iint_{\partial V} \mathbf{F} \cdot d\mathbf{S} = \iiint_V (\nabla \cdot \mathbf{F}) dV.$$

Converts flux through a closed surface to a volume integral of the divergence.

Theorem 14.2 (Stokes' Theorem).

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}.$$

Note: Green's theorem is the 2D special case.

15 Differential Forms & Generalized Stokes (Week 12)

Definition 15.1 (Differential forms). • **0-form:** function f .

- **1-form:** $\omega = P dx + Q dy + R dz$.
- **2-form:** $\omega = P dy \wedge dz + Q dz \wedge dx + R dx \wedge dy$.
- **3-form:** $\omega = f dx \wedge dy \wedge dz$.

Definition 15.2 (Exterior derivative). d maps k -forms to $(k+1)$ -forms: $d(0\text{-form}) = \text{gradient}$, $d(1\text{-form}) = \text{curl}$, $d(2\text{-form}) = \text{divergence}$.

Key property: $d(d\omega) = 0$.

Theorem 15.3 (Generalized Stokes' Theorem).

$$\int_{\partial\Omega} \omega = \int_{\Omega} d\omega.$$

<i>Theorem</i>	<i>Dimension</i>
<i>Fundamental Theorem of Calculus</i>	<i>1D</i>
<i>Green's Theorem</i>	<i>2D</i>
<i>Classical Stokes' Theorem</i>	<i>3D (curve $\leftrightarrow$ surface)</i>
<i>Gauss / Divergence Theorem</i>	<i>3D (surface $\leftrightarrow$ volume)</i>

Problem-Solving Strategy Cheat Sheet

1. **Improper integral convergence?** Identify type (I or II). Find asymptotic equivalent near the singular point. Compare with p -integral.
2. **Multivariable limit?** Try two paths first (to disprove). If paths agree, try polar/squeeze (to prove).
3. **Continuity at $(0,0)$?** Check $\lim f = f(0,0)$ using polar or squeeze.
4. **Differentiability at $(0,0)$?** (1) Compute $f_x(0,0)$, $f_y(0,0)$ from definition. (2) Verify $\lim_{\Delta\rho \rightarrow 0} [\Delta f - f_x \Delta x - f_y \Delta y] / \Delta\rho = 0$. Or: if f_x, f_y are continuous $\Rightarrow$ differentiable (Thm 3).
5. **Tangent plane?** For $z = f$: use gradient. For $F = C$: use ∇F .
6. **Chain rule?** Draw dependency tree; multiply partial derivatives along each path and sum.
7. **Local extrema?** Solve $\nabla f = 0$, apply Hessian/Sylvester.
8. **Constrained optimization?** Lagrange multipliers.
9. **Global extrema on compact domain?** Critical points inside + boundary extrema; compare all values.
10. **Double integral?** Sketch region; choose order of integration; use polar if $x^2 + y^2$ appears.

11. **Triple integral?** Cylindrical if $x^2 + y^2$; spherical if $x^2 + y^2 + z^2$.
12. **Line integral?** Type I: parametrize + $|\mathbf{r}'|$. Type II: check conservative first $\rightarrow$ potential or Green's.
13. **Surface integral / flux?** Parametrize + cross product. Closed surface $\rightarrow$ Gauss.
14. **Closed curve in 3D?** Consider Stokes' theorem.